PROJECT REPORT

SPAM MESSAGE DETECTOR

WINTER SEMESTER 2025-2026

Name: Priya Ranjan

Registration Number: 25BCE10221

Course Name: Fundamentals of AI and ML

Course Code: CSA2001

Slot: A14+D11+D12

Faculty: Prof. Suneet Joshi

Class Number: BL2025260500808

INTRODUCTION

In today's advance growing world where technology is updating so fast and so are we and so are crimes. Most of the work we do today are online. Our most of the time is being spent online. So, one of the issues that we face online is **"SPAM MESSAGES."** Spam messages are those unwanted and risky messages that are risky for users and usually scams. The use of digital communication is increasing so fast that now understanding what spam messages are it is very important.

The project that I have made **"SPAM MESSAGE DETECTOR**" uses machine learning. The project that I have made helps the users to identify messages into Spam and Not Spam (Ham). It helps the users to know spam messages and tell them to whether trust it or not.

Problem Statement

The problem I chose is **"SPAM MESSAGE DETECTOR."** I chose this problem because we really cannot find the difference between spam messages anymore. It is very difficult to decided which message is spam and which is not because of how we text and talk to people. The main thing this project does is that to make a system that detects spam messages automatically using machine learning.

We often reply to texts fastly. We not make a decision before replying it. If it is a scam message, you face the consequence later. Making decisions and replying is time-consuming and often we reply without thinking much. So, an automated system will help us as it will detect messages nicely.

OBJECTIVES

- Building a machine learning model for detecting spam messages.
- To convert text into numerical using TF-IDF
- To train classifier using Naïve Bayes Algorithm
- It develops both- Command Line Interface (CIL), Graphical User Interface (GUI)

FUNCTIONAL REQUIREMENTS

- The system allows the user to enter the messages.
- It pre-processes the text.
- It converts the text into numerical using TF-IDF.
- It classifies the messages using Naïve Bayes Algorithm.
- It shows the results as- Spam or Not Spam
- It shows both versions- Command Line Interface (CLI) and Graphical User Interface (GUI)
- It loads the training model (model.pkl) to predict
- It handles multiple inputs and you can type exit to stop in CLI version.

NON-FUNCTIONAL REQUIREMENTS

- **Performance:** It tells the system fastly whether the message is spam or not.
- **Usability:** It is easy to use for anyone.
- **Reliability:** It produces accurate results.
- **Scalability:** It can handle many inputs.
- **Efficiency:** It uses very less memory and resources.

DATASET USED

- The dataset which I have used here is SMS Spam Collection Dataset.
- It contains two messages: Spam and Not Spam
- I have stored it in the file named spam.csv.
- It includes the messages that we use in our world for testing and training.

DATASET LINK:

https://www.kaggle.com/datasets/uciml/sms-spam-collection-dataset

METHODOLGY USED

1. **Data processing:**
    - It loads various dataset using pandas

2. **Feature Extraction:** It converts text into numerical format using TF-IDF which helps the model understand the importance of words in messages.

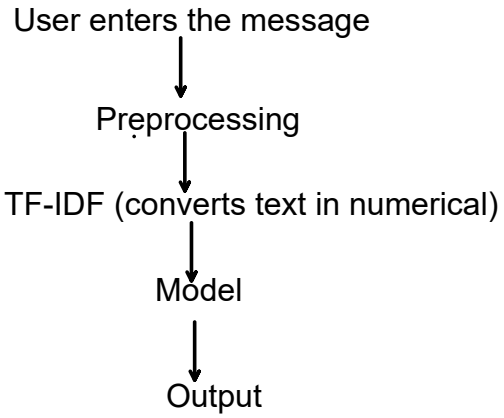
3. **Model Training:** The algorithm used here is Naïve Bayes Algorithm. Dataset is made of two different things that is, training and testing and the model learn patterns from training data.

4. **Model Evaluation:** The accuracy is calculated and the model testing is based on non-written data. Performance is being judged.

5. **Model Saving:** The training model is saved as model.pkl whereas the vectorizer is saved as vectorizer.pkl.

6. **Prediction System:** It displays the result in two interfaces. One is Command Line Interface (CLI) where the user enters the message in the terminal and it tells spam or not. The other is Graphical User Interface (GUI) where the user enters the message in the GUI window and it checks whether the message is spam or not.

SYSTEM ARCHITECTURE



TECHNOLOGIES USED

- Programming language: Python
- Libraries used:
    1. pandas
    2. scikit-learn
    3. joblib
    4. tinkter

IMPLEMENTATION

- train.py: it is used for training the model.
- predict.py: it is used for the CLI result.
- gui.py: it is used for the GUI result.
- model.pkl: it is for saved model.
- vectorizer.pkl: it is for saved vectorizer.
- spam.csv: it is the dataset of spam messages.

KEY DECISIONS MADE

- I took the use of Naïve Bayes Algorithm for machine learning technique because I wanted faster and correct results.
- I have used TF-IDF for converting my text into numerical because it easily helps me detect words in my spam message detector.
- I have used SMS Spam Collection Dataset because it is used everywhere and has every kind of message that can be considered as scam.
- I have used CLI and GUI versions both because it made the project look easier. CLI version can be executed multiple times and helps in testing whereas GUI for user-friendly interaction.
- I have made a simple model because I wanted faster training and fast results and is easily understood by beginners.
- The model and vectorizer both are saved so it has not to go back every time it runs.

SCREENSHOT of RESULT:

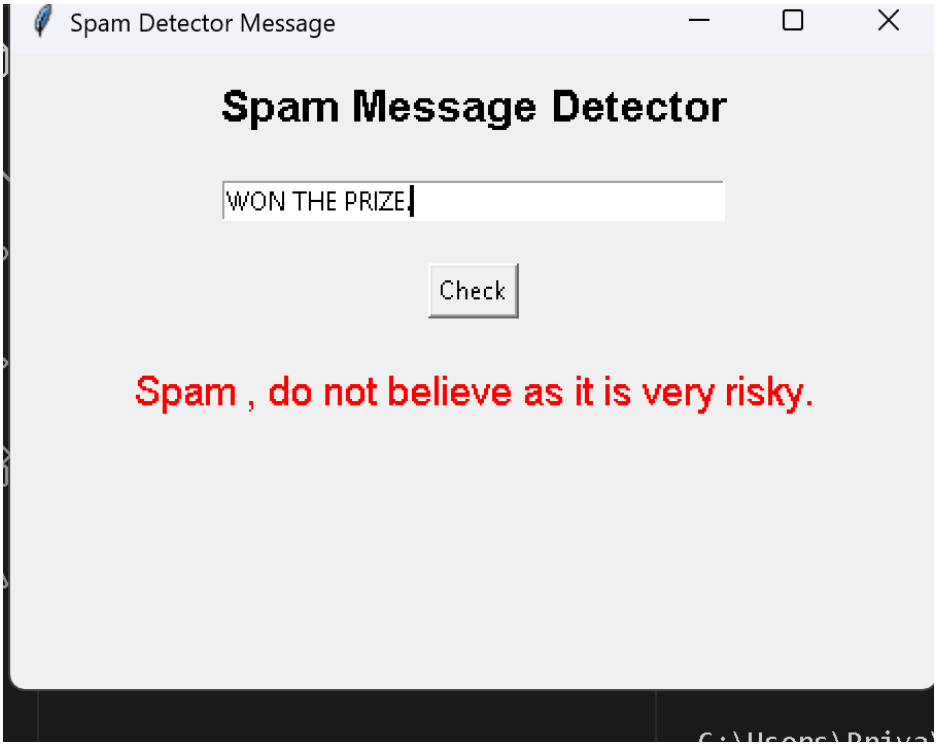
1. CLI INTERFACE (train.py)

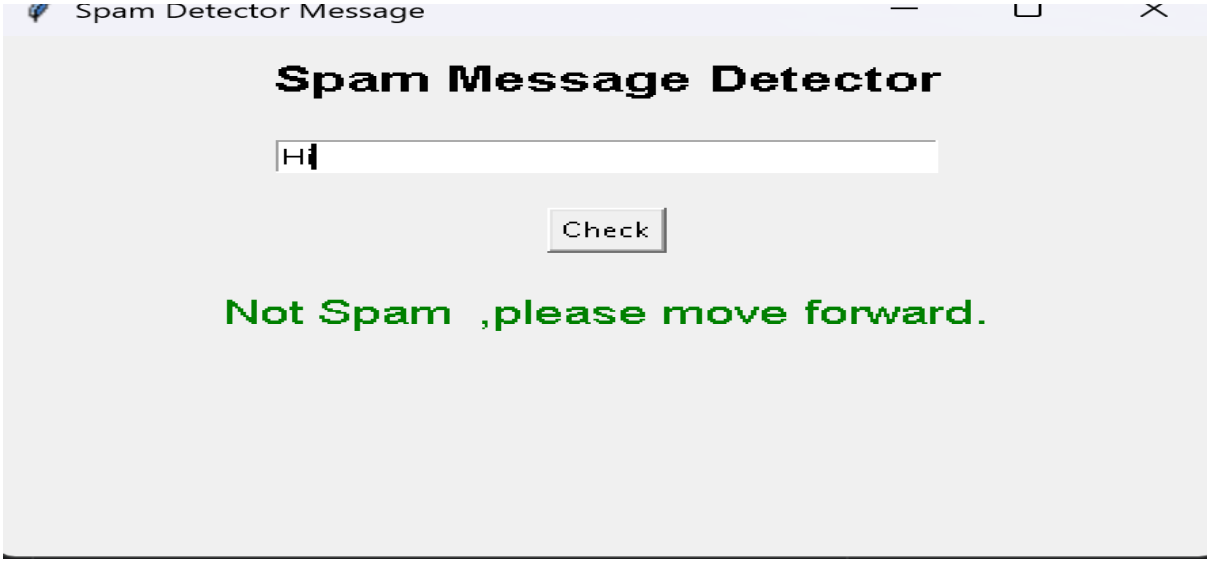
```
PROBLEMS    OUTPUT    TERMINAL    ...                                cmd

Microsoft Windows [Version 10.0.26200.8037]
(c) Microsoft Corporation. All rights reserved.

C:\Users\Priya\Downloads\spamdetector> python train.py
Accuracy: 0.9013452914798207

Classification Report:

              precision    recall  f1-score   support

           0       0.99      0.89      0.94       965
           1       0.58      0.95      0.72       150

    accuracy                           0.90      1115
   macro avg       0.79      0.92      0.83      1115
weighted avg       0.94      0.90      0.91      1115


Saved: model.pkl, vectorizer.pkl

C:\Users\Priya\Downloads\spamdetector>
                                   Ln 62, Col 17   Spaces: 4   UTF-8   LF
```

2. CLI INTERFACE (predict.py)

```
PROBLEMS    OUTPUT    TERMINAL    ...                                cmd

C:\Users\Priya\Downloads\spamdetector> python predict.py
Spam Messagr detector is working , just a little time.  (type 'exit' to st
op whenever you want)

Enter message: :hi.
Not Spam, you can go forward.

Enter message: WIN
Spam, risky do not believe this message.

Enter message: WIN PRIZE.
Spam, risky do not believe this message.

Enter message: "HI."
Not Spam, you can go forward.

Enter message: exit
I was glad to help you. Byee !!

C:\Users\Priya\Downloads\spamdetector>
```

3. GUI INTERFACE





CHALLENGES FACED

- It was difficult to first handle text data preprocessing.
- It was difficult to understand how text was converted to numerical.
- I had issues in fixing my environment and installing of libraries.
- I had a trouble to make my CLI interface keep executing.
- I had to search a lot for a good spam message dataset.
- It was a trouble to debug the file path and module errors.

KEY LEARNINGS

- I have understood about machine learning basics and it had helped me to understand how machine learning based model works and how to classify tasks.
- It has helped me understand about TF-IDF and how by using it my text converts to numerical.
- It has also helped me use a model (Naïve Bayes Algorithm) which helped a lot in my project.
- It has helped me about library such as pandas which was used in my model building.
- I have also learned on how to save training models.
- I have also learned using both CLI interface and GUI interface.

FUTURE SCOPE

- I want to use more deep learning models to make it more accurate.
- I want to improve my GUI design.
- I want to also convert into web application.
- I want to add real-time message filtering

REFRENCES

- Vityarthi lessons: Fundamentals in AI and ML
- SMS Spam Collection Dataset (Kaggle)
- TF-IDF for converting text into numerical
- Python libraries such as pandas, scikit-learn, joblib and tkinter
- Naïve Bayes Algorithm Tutorial
- Online resources for GUI window

GITHUB LINK-

https://github.com/priya25bce/_spam_message_detector